

Toward explainable electrical load forecasting of buildings: A comparative study of tree-based ensemble methods with Shapley values

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ABSTRACT

Electrical load forecasting of buildings is crucial in designing an energy operation strategy for smart city realization. Although artificial intelligence techniques have demonstrated excellent energy forecasting performance, explaining how outcomes are obtained when they are inaccurate is challenging. Explainable artificial intelligence (XAI) has recently received considerable attention in addressing this issue. This study proposes an explainable electrical load forecasting (XELF) methodology. We first preprocess data for input variable configuration and build the following tree-based ensemble models that produce outstanding results in tabular data: random forest, gradient boosting machine (GBM), extreme gradient boosting, light GBM (LightGBM), and categorical boosting. We evaluate performance in terms of the mean absolute percentage error, coefficient of variation of the root mean square error, and normalized mean absolute error. Finally, we provide the rationale for interpreting the influence of the input variables and decision-making process via Shapley additive explanations (SHAP), an XAI technique, on the best model. The experiments were conducted with an electrical load dataset from educational buildings to validate the practicality and validity of this methodology. We applied the SHAP to the LightGBM model and performed its respective analyses and visualizations for XELF.

Introduction

The rise in the urban population has resulted in many issues, including resource scarcity, traffic congestion, and greenhouse gas (GHG) emissions [1,2]. Many towns and governments have been working hard to transition to smart cities to operate complex urban systems effectively [2,3]. The smart city concept is intended to increase the efficiency and security of urban infrastructures while improving the quality of life [4]. For example, smart cities can cut GHG emissions by lowering traffic congestion and energy consumption and introducing alternatives, such as electric vehicles, energy storage systems (ESSs), and renewable energy sources [5,6]. Building energy efficiency, mainly through energy storage systems and renewable energy sources, is crucial in cities because building energy use is a significant source of GHG emissions [7]. Therefore, most smart city systems have been developed to implement a variety of city services, such as energy management systems (EMSs), that can improve the energy efficiency of the existing buildings [8].

An EMS is a computer-aided tool that efficiently controls and

monitors energy consumption using the Internet of Things (IoT) and big data technologies [9,10]. An EMS can be classified into a building EMS (BEMS), factory EMS, home EMS, or cluster or community EMS according to the purpose of the building [11]. The EMS can provide the maximum economic benefit by trading surplus energy or preventing blackouts by reducing daily peak demand in a building based on short-term load forecasting (STLF) [12]. However, accurate STLF is challenging because building energy consumption has diverse patterns and considers a nonlinear relationship with external factors [12,13]. Many studies have conducted STLF based on artificial intelligence (AI) techniques because they can effectively train recent energy consumption patterns and the nonlinear relationship between the input and output variables [14,15].

Although expectations regarding AI technologies in industrial fields have also been growing due to their excellent performance, their application is not easy [16,17]. In addition, AI technology consists of a longitudinal learning structure that derives results from the given data [18]. Most methods that have excellent performance present a form in which it is difficult to confirm the internal decision-making process, similar to a black-box state [16,19]. For example, an STLF model based

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Abbreviations

AI	artificial intelligence	LightGBM	light gradient boosting machine
Bagging	bootstrap aggregating	LIME	local interpretable model-agnostic explanation
BEMS	building energy management system	LSTM	long short-term memory
CatBoost	categorical boosting	MAPE	mean absolute percentage error
COVID-19	coronavirus disease 2019	MLP	multilayer perceptron
CVRMSE	coefficient of the variation of the root mean square error	MLR	multiple linear regression
DNN	deep neural network	NMAE	normalized mean absolute error
DOTW	days of the week	PDP	partial dependence plot
DT	decision tree	RF	random forest
ELI5	explain like I'm five	SHAP	Shapley additive explanations
EMS	energy management system	STLF	short-term load forecasting
ESS	energy storage system	SVR	support vector regression
GBM	gradient boosting machine	THI	temperature-humidity index
GHG	greenhouse gas	WCT	wind chill index
IoT	Internet of Things	XAI	explainable artificial intelligence
KMA	Korea Meteorological Administration	XELF	explainable electrical load forecasting
		XGBoost	extreme gradient boosting

on a famous AI technique that usually has excellent prediction performance can yield a wildly inaccurate prediction at a specific point. If the EMS manager cannot explain how the predictions of this model are reached, this model would be challenging to use for the EMS. Recently, explainable AI (XAI) techniques have attracted considerable attention in addressing this issue [20,21].

Moreover, XAI is a research field that can allow users to explain the process of generating outcomes by interpreting the AI technology training process and prediction results [21–23]. It can change the decision-making process inside AI to secure reliability, provide stability, and improve model performance [22,23]. The research directions for XAI can be divided into two broad categories [23]. The first approach is to propose a novel AI-based model that can demonstrate the decision-making process while maintaining performance similar to the targeted AI-based model. The second approach is to add a descriptive label to the part that generates the result inside the existing AI-based model and transform it so that users can confirm how the results are derived. Because decision tree (DT)-based ensemble learning models have recently achieved excellent performance in many fields, research corresponding to the second approach has been actively conducted [24–26].

Because STLF is a core technology for implementing the EMS, accurate STLF results and an explanation of how predictions are derived are required for an EMS to operate stably and efficiently. For example, when an EMS manager issues a peak alert to a customer based on a forecasted load, if the manager explains how the forecasted load was derived, the customer can easily be persuaded and actively respond to the peak. Electric utilities are eager to build next-generation control centers with visualization and business analytics tools that support emerging technologies, such as AI and mixed reality; however, they also aim to make the usability of these technologies easier for employees with less experience with them [27]. The STLF systems based on XAI and comparable tools provide a more complete basis for electric utilities. Although many studies have been conducted using XAI technologies for energy data analytics, limitations in electrical load forecasting persist [27–30].

This paper presents an explainable electrical load forecasting (XELF) model for educational buildings. We build an STLF model using five DT-based ensemble learning methods: random forest (RF), gradient boosting machine (GBM), extreme gradient boosting (XGBoost), light gradient boosting machine (LightGBM), and categorical boosting (CatBoost). Then, we compare the performance in terms of the mean absolute percentage error (MAPE), coefficient of the variation of the root mean square error (CVRMSE), and normalized mean absolute error

(NMAE) and use Shapley additive explanations (SHAP), an XAI technique, to explain the best DT-based ensemble learning model [31]. This study can aid utility engineers and researchers working on power generation and load forecasting by offering insight into the XAI potential and availability in smart grid applications.

The rest of this paper is structured as follows. Section 2 summarizes recent work on XAI for energy data analysis and STLF for educational buildings. Section 3 describes data preprocessing to configure input variables for training DT-based ensemble learning models. Section 4 details the development of the XELF model. Section 5 presents and discusses the main findings of the experiments. Finally, Section 6 provides the conclusions and future research directions.

Related work

Several studies have been undertaken on forecasting models for building electrical energy consumption and XAI methodologies for energy data analytics based on AI approaches, such as DT-based ensemble learning and neural network-based deep learning methods. We introduce the application case of the XAI technologies for energy data analytics and recent STLF approaches in educational buildings.

XAI for energy data analytics

Kuzlu et al. used three XAI tools – local interpretable model-agnostic explanation (LIME), SHAP, and explain like I'm five (ELI5) – for explainable solar photovoltaic forecasting [27]. They constructed an RF-based forecasting model for hourly photovoltaic energy forecasting. Then, they applied XAI tools to the RF model to elucidate the reasons for the results. They also examined the strengths and limitations by considering the computing cost, feature weight, explanation locally and globally, and other factors. Park et al. proposed an explanation of an anomaly detection scheme for a differential pressure control valve in district heating [28]. They constructed an RF-based anomaly detection model and used SHAP to present the contribution of input variables to the derived outcome. They confirmed that the RF model performed better than the logistic regression and DT models and that the value of a particular sensor was the most crucial variable. In addition, the contribution of that variable was shown through their experiments. Chakraborty et al. developed an explainable scenario-based building cooling energy consumption forecasting scheme using EnergyPlus [29]. They constructed an XGBoost-based forecasting model by considering future climate change projections and presented the local explanations of the XGBoost model, demonstrating how cooling load varies with the

temperature-driven climatic conditions and building types based on SHAP. Amiri et al. developed an explainable forecasting approach for household transportation energy consumption using the Delaware Valley Regional Planning Commission open-source Household Travel Survey [30]. They constructed a deep neural network (DNN) model to predict household transportation energy consumption and applied LIME to the DNN model to understand model decisions.

STLF for educational buildings

Son et al. developed a 15-minute-interval electricity consumption forecasting model for a private university in Seoul, South Korea [32]. They used a deep autoencoder for feature extraction and RF for prediction. The proposed model achieved better performance than the original and principal component analysis for feature extraction and support vector regression (SVR) and multilayer perceptron (MLP) for prediction. Ruiz-Abellón et al. proposed four 48-hour-ahead STLF models for educational buildings in Spain based on bootstrap aggregating (Bagging), RF, conditional RF, and XGBoost [33]. Their experiments indicated that the XGBoost model yielded a better prediction than the other STLF models and verified that the historical load one week prior was the most significant variable in constructing the XGBoost model. Wang et al. proposed an RF-based hourly electricity consumption forecasting model for two educational buildings located at the University of Florida [34]. They compared the performance of the RF model with that of the DT and SVR models to verify the superiority of the RF model. They also constructed an RF model for each semester and confirmed the input variables most associated with electricity consumption by semester via the variable importance functions. Bellahsen and Dagdougui performed STLF of 15-minute-, 2-hour-, and 24-hour-ahead intervals on five buildings collected from the BEMS of an urban educational district in Montréal, Canada [35]. They used the MLP, long short-term memory (LSTM) network, sequence-to-sequence LSTM network, k-nearest neighbors, and RF for the STLF model construction. They considered district- and building-level forecasting approaches and confirmed that the RF model exhibited better performance than the other STLF models.

Input variable configuration

This section introduces the input variable configuration process for constructing the STLF model. We focused on day-ahead hourly electrical load forecasting because the hour is the period widely used for optimal power system operation from one day later, such as demand response, ESS scheduling, and renewable electricity generation [36–38]. We collected educational building electricity consumption data from Sejong University [39].

The educational institution employs the Power Planner, a service application to provide electricity-related statistics to use electricity more efficiently, such as estimated bills, electricity consumption, and load pattern analysis results. We collected hourly electricity consumption data for five years through the Power Planner from March 2015 to February 2021. The data contained missing values at 275 time points (approximately 0.006 %). We replaced the missing values via our previous study for missing electricity consumption data imputation [10].

Timestamp information

We considered external and internal factors for the input variable configuration. Typical external factors include timestamp and weather information [40,41]. A variable primarily considered in internal factors is historical electricity consumption [42,43].

The timestamp information, such as months, hours, days of the week, and holiday indicators, can present different electricity consumption patterns for buildings [9]. For instance, hourly electricity consumption patterns differ depending on working hours, mealtimes, and other

factors. The days of the week and holiday indicators can also present different electricity consumption patterns depending on working days (e.g., Monday to Friday) and weekends (e.g., Saturday and Sunday). Thus, because the days of the week and hours have a sequence format, we used Equations (1) to (2) to enhance the categorical variable in the one-dimensional space to the continuous variable in the two-dimensional space to effectively reflect the periodicity of these variables [9,41]:

$$Time_x = \sin(360^\circ / Period_{Time} \times Time), \quad (1)$$

$$Time_y = \cos(360^\circ / Period_{Time} \times Time), \quad (2)$$

where *Time* is the time unit, we want to transform to reflect the periodicity, namely, *Hour* and *DOTW*, which denote the hours of the day and days of the week, respectively, and *Period_{Time}* is its period. For example, *Period_{Time}* will be 24 if we convert the hours of the day, while *Period_{Time}* will be seven if we convert the days of the week.

In addition, to reflect the different electricity consumption patterns on weekdays and holidays, we considered holiday indicators to be an input variable. We added the holiday indicators for the educational building dataset, including weekends (Saturday and Sunday) and national holidays, as a binary variable (1: corresponding to the date, 0: otherwise).

Weather information

Weather information is one of the most significant factors in STLF because the frequency of use of high energy-consuming appliances, such as air conditioners or heaters, is closely related to weather [44,45]. The Korea Meteorological Administration (KMA) provides various types of short-term weather forecast information for every region in South Korea: weather observation, probability of precipitation, amount of precipitation, daily maximum/minimum temperatures, temperature, wind speed/direction, and humidity [45].

To increase the applicability of the proposed method for real-world applications, we used temperature, humidity, and wind speed as input variables because they are closely related to electricity consumption [12,13]. We used the KMA automated synoptic observing system data, collected by the Seoul Meteorological Observatory, located about 10 km from the Sejong University campus, as the weather factors for the educational building.

We calculated the temperature-humidity index (THI; also known as the discomfort index) and wind chill temperature (WCT) to establish a more direct association with electricity consumption (i.e., how warm one feels in summer or how cold one feels in winter) using Equations (3) and (4), respectively [45]. The temperature, humidity, and wind speed are denoted by *Temp*, *Humi*, and *WS*, respectively. As a result, we considered 10 external factors as input variables for model training.

$$THI = (1.8 \times Temp + 32) - [(0.55 - 0.0055 \times Humi) \times (1.8 \times Temp - 26)], \quad (3)$$

$$WCT = 13.12 + 0.6215 \times Temp - 11.37 \times WS^{0.16} + 0.3965 \times Temp \times WS^{0.16}. \quad (4)$$

Historical electricity consumption

We considered historical electricity consumption data to be internal factors because they present recent electricity consumption trends [12,46]. To that end, we used electricity consumptions of one day and one week prior to the prediction time point [12]. The electricity consumption one day prior can indicate the most recent trends for the hour of the day, and the electricity consumption one week prior can indicate the most recent trends for the day of the week [9,19]. Because the electricity consumption patterns of weekdays and holidays can differ,

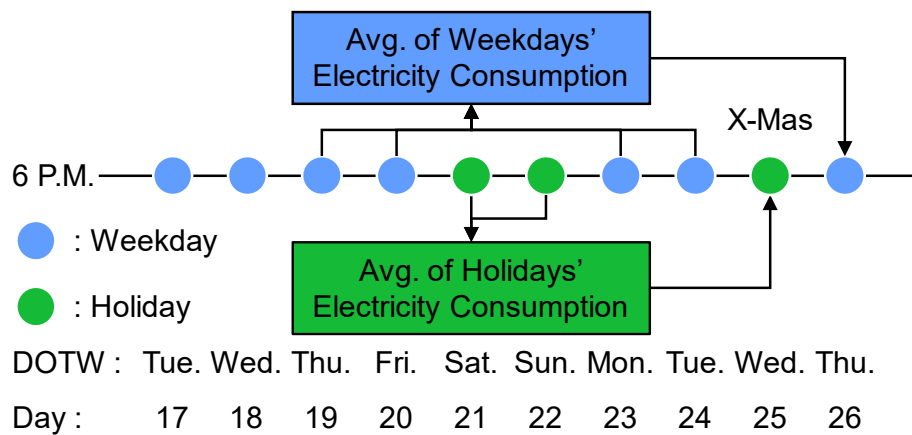


Fig. 1. Example of the weekly average electricity consumption calculation (December 2019). DOTW: days of the week; X-Mas: Christmas.

Table 1
Input variables and their information for XELF modeling.

Timestamp information	Description	Weather information	Description	Historical electricity consumption	Description
$Hour_x$	Sine value on the hour (numeric)	$Temp$	Temperature (numeric)	$Cons_1$	Electricity consumption one day prior (numeric)
$Hour_y$	Cosine value on the hour (numeric)	$Humi$	Humidity (numeric)	$Holi_1$	Holiday indicator one day prior (binary)
$DOTW_x$	Sine value on the day of the week (numeric)	WS	Wind speed (numeric)	$Cons_7$	Electricity consumption one week prior (numeric)
$DOTW_y$	Cosine value on the day of the week (numeric)	THI	Temperature-humidity index (numeric)	$Holi_7$	Holiday indicator one week prior (binary)
$Holi$	Holiday indicator (binary)	WCT	Wind chill temperature (numeric)	$Cons_{avg}$	Weekly average electricity consumption (numeric)

we also added holiday indicators to indicate whether the day was a holiday for both electricity consumption types [19].

In addition, we added an innovative historical electricity consumption value as an internal factor to effectively reflect the electricity consumption trend for the prediction time point during a week [9,46]. For this, we considered two cases. First, when the prediction time point was a weekday, we calculated the average of the weekday electricity consumption over the preceding seven days. Second, when the prediction time point was a holiday, we calculated the average of the holiday electricity consumption over the preceding seven days.

Fig. 1 depicts an example of computing the average electricity consumption on December 25 (Christmas) and 26, 2019. For example, for a prediction time point on a holiday, such as Wednesday, December 25, 2019, at 6 p.m., we calculated the average power consumption on Saturday, December 21, 2019, at 6 p.m. and Sunday, December 22, 2019, at 6 p.m. If the prediction time point was on a weekday, such as Thursday, December 26, 2019, at 6 p.m., the average was calculated from the electricity consumption at 6 p.m. from Thursday, December 19; Friday, December 20; Monday, December 23; and Tuesday, December 24, 2019. As a result, we considered five internal factors as input variables for model training and presented all input variables and their information in [Table 1](#).

XELF model development

This paper uses an ensemble learning technique that achieves excellent performance in a tabular form for the XELF model construction [47–49]. We first divided each preprocessed dataset of 15 input variables into training and testing sets in an 80:20 ratio. We used a grid search through fivefold cross-validation to select the optimal hyperparameter values for each DT-based ensemble learning model in the training set. Then, we built the DT-based ensemble models with the best

hyperparameter values. In the testing set, we compared the prediction performance of the DT-based ensemble learning models to verify the best STL model. Finally, we applied the SHAP to the best STL model using the training set to interpret the influence of input variables and the decision-making process.

STLF model construction

Ensemble learning is a methodology that strategically creates and combines multiple models to derive better classification or prediction performance than a single model [50]. Typical ensemble learning techniques include bagging, boosting, and stacking, and we considered RF for the bagging approach and GBM, XGBoost, LightGBM, and CatBoost for the boosting approach [49,50]. Because the DT and DT-based ensemble learning methods divide the entire set of observations at each node by the value of one of the features, they are not sensitive to the data variance [51,52]. Therefore, DT-based ensemble learning methods do not require feature scaling. Thus, we did not perform feature scaling techniques, such as min-max normalization and standardization.

Bagging is a relatively straightforward technique that can decrease the result variance [50,53]. Bagging performs random sampling of a given dataset and constructs a weak (single) model from each sampled dataset. Then, bagging aggregates the prediction values of the weak models to derive an outcome via voting or averaging for classification or regression, respectively. A DT, a high-variance machine learning method, is generally used as a weak model in the bootstrap procedure [53]. In addition, RF, one of the most popular bagging methods, works by constructing several weak DTs trained on a random subset of variables and delivers an outcome by considering all the DTs [54,55]. Moreover, RF can effectively process massive datasets and has excellent performance in regression and classification tasks because it addresses numerous independent variables without removing them [55].

Boosting is similar to bagging in that it aggregates an outcome but adopts a different learning strategy from bagging [56]. Boosting generates new weak models sequentially for ensemble learning (i.e., it trains a new weak model, such as DT, by considering the whole ensemble error trained so far in each specific iteration). The GBM is based on a gradient descent algorithm that adjusts parameters iteratively to decrease the loss (cost) function [56,57]. The GBM evaluates the local gradient of the loss function for a given dataset of parameters and performs steps in the descending gradient direction. An essential parameter in the gradient descent is the learning rate, which can control the step size. When the GBM is constructed with an increasingly low learning rate and many DTs, it is computationally expensive (e.g., time-consuming) and can lead to overfitting due to a sequential learning procedure.

Chen and Guestrin developed XGBoost to implement an efficient and scalable GBM [58]. To handle the time-consuming problem of GBM, XGBoost produces more accelerated training and scalability based on distributed and parallel computing by applying the column subsampling method [58,59]. Further, XGBoost can efficiently handle missing values because these values are automatically supplemented and recognized to achieve boosting [58]. Moreover, XGBoost increases horizontally by first passing through nodes close to the root node to reduce and balance the tree depth. Furthermore, to avoid overfitting, XGBoost sets an objective function at iteration by appending the regularization to the loss function.

The LightGBM, released by Microsoft in 2016, is a machine learning technique based on gradient boosting DTs [60]. Such techniques (e.g., GBM and XGBoost) are limited because they do not perform well on numerous high-dimensional data, as all data must be scanned to calculate the information gain for all possible split points. The LightGBM uses the gradient-based one-side sample, which gathers information by removing the data portion with a slight slope, and exclusive feature bundling techniques to overcome this problem [59,60]. Because data with a significant slope are critical in obtaining information, information can be approximated quickly and reliably even with minimal data, eliminating the slight slope. Exclusive feature bundling is a strategy that groups and processes mutually exclusive variables to reduce the number of variables efficiently without considerably compromising the split-point determination accuracy by grouping and processing variables that infrequently simultaneously have nonzero values. The LightGBM model has benefits, such as lower memory utilization and faster training speed, using these techniques [61].

Yandex developed CatBoost, a novel GBM approach, to efficiently handle categorical variables [62]. Moreover, CatBoost is an ordered boosting method that focuses on categorical variable preprocessing and overfitting problems. Unlike the existing boosting model, which learns all residual errors sequentially, ordered boosting builds a model by calculating the residual error of some data and calculating the residual error of the remaining data using that model [62,63]. The CatBoost algorithm enhances training speed by grouping variables with the same information gain into feature combinations. It also inhibits overfitting by random permutation for the order of data in ordered boosting. The CatBoost algorithm calculates the average sample value of variables with the same category from a dataset that has undergone random permutation for categorical variable preprocessing.

Model explainability with SHAP

In the case of a sophisticated AI-based model, it is difficult to determine how much each input variable influences the outcome [22,28]. As a result, a strategy for approximating the anticipated values of the original model to an interpretable model is required [22]. Further, SHAP is a tool that uses the Shapley value to examine the correlation between independent and dependent variables in model learning [31]. The Shapley value in game theory indicates that all advantages acquired from game players are dispersed based on each player's contribution [21,26]. For example, in the variable importance of the DT-based

ensemble learning model, the game is the expected value of the observation, the participant is the independent variable, and the contribution is variable importance. The Shapley value follows Equation (5) [16,21]:

$$\Phi_i = \sum_{S \subseteq F \setminus \{i\}} (|S|! \times (|F| - |S| - 1)!) / (|F|!) \times [f_{S \cup \{i\}}(x_{S \cup \{i\}}) - f_S(x_S)], \quad (5)$$

where F is the set of all independent variables, f is the learning model, and i is the independent variable. In addition, Φ_i is the contribution with respect to the set, S , of all independent variables that satisfy $S \subseteq F$, $f_{S \cup \{i\}}(x_{S \cup \{i\}}) - f_S(x_S)$, which is the difference between the presence and absence of the corresponding independent variable, i . The contribution for each subset, S , $|F|!$ (number of cases in which all sets F are arranged in a row) is a weighted average with $|S|!$ (number of cases in which S is arranged in a row) and $(|F| - |S| - 1)!$ (number of cases in which everything except i and S is arranged in a row).

Moreover, SHAP determines the Shapley value of the learning model using the conditional expected value function and includes the kernel SHAP (applicable to all machine learning models), deep SHAP (applicable to neural network models), and tree SHAP (applicable to DT-based ensemble learning models) [25,31]. In this paper, we employed tree SHAP to compute the SHAP values for all independent variables for each dataset tuple [16,17]. This method allows us to represent the variable importance and partial dependence plots (PDPs). The average sum of the SHAP values or absolute values of SHAP can visualize the variable importance [25]. For all tuples, the PDPs represent the input value of the corresponding independent variable and SHAP value as a dot.

The existing feature importance technique uses the permutation method to measure the influence of a variable in a model [19]. Although this method has the advantage of fast computation, the outcome may be misinterpreted when the variables are dependent on each other. Furthermore, the negative influence cannot be calculated. As a result, the value of a given variable may be greater than its actual influence. However, the SHAP method considers the likelihood that variables influence one other and can quantify the negative influence [17,24]. Hence, despite its disadvantage of being slow, it assesses influence more precisely than the variable importance method.

Experimental results

Experimental design

This section describes the experiments and compares the performance of DT-based ensemble models. For performance comparison, we use three metrics: MAPE, CVRMSE, and NMAE. Because these metrics represent accuracy as a percentage error, they are more intuitive and understandable [45,46]. Equations (6) to (8) calculate MAPE, CVRMSE, and NMAE, respectively:

$$MAPE = 1/n \times \sum |(Y_t - \hat{Y}_t)/Y_t| \times 100, \quad (6)$$

$$CVRMSE = \sqrt{1/n \times \sum (Y_t - \hat{Y}_t)^2} / \bar{Y} \times 100 \quad (7)$$

$$NMAE = (1/n \times \sum |Y_t - \hat{Y}_t|) / \bar{Y} \times 100, \quad (8)$$

where n is the number of prediction time points, t denotes the prediction time point, Y_t and $\hat{Y}_t$ represent the actual and predicted values at time t , respectively, and $\bar{Y}$ indicates the average of all actual values.

We roughly separated the datasets into training (in-sample) and testing (out-of-sample) sets in the ratio of 80:20. We used hourly electricity consumption data from March 1 to 7, 2015, in the educational building dataset to configure input variables for the training set of the DT-based ensemble learning models. The educational building dataset from March 8, 2015, to February 28, 2019, and March 1, 2019, to

Table 2
List of hyperparameters for tree-based ensemble learning models.

Models	References	Hyperparameters
RF	[64]	Number of trees: 128
GBM	[59]	Number of features: <i>auto</i> , <i>sqrt</i> , <i>log2</i> Learning rate: 0.01, 0.05, 0.1
XGBoost	[59]	Number of iterations: 100, 250, 500 Maximum depth of the tree: 5, 10 Loss function: quantile regression, Huber loss Learning rate: 0.01, 0.05, 0.1
LightGBM	[59]	Number of iterations: 250, 500, 1000 Maximum depth of the tree: 6, 8, 10 Subsample: 0.5, 0.75, 1.0 Colsample by tree: 0.5, 0.75, 1.0 Colsample by level: 0.5, 0.75, 1.0 Colsample by node: 0.5, 0.75, 1.0 Tree booster: <i>gbdt</i> , <i>dart</i> Learning rate: 0.01, 0.05, 0.1
CatBoost	[65]	Number of iterations: 1000, 1500 Number of leaves: 64 Subsample: 0.5 Colsample by tree: 1.0 Tree booster: <i>gbdt</i> , <i>dart</i> Learning rate: 0.03, 0.1 Maximum depth of the tree: 4, 6, 10 L2 regularization: 1, 3, 5, 7, 9

February 29, 2020, were used as the training and testing sets, respectively.

The hyperparameters of most AI approaches can influence model performance. The hyperparameters must be fine-tuned to boost their performance even further. To discover the best hyperparameters, we used GridSearchCV in scikit-learn to determine the best values for the hyperparameters of DT-based ensemble learning models. Table 2 lists the hyperparameters for each model and the references that provide information on their respective effective values or hyperparameter ranges.

Table 3
Performance comparison of tree-based ensemble models trained with external factors for the educational building dataset. W, H, and T indicate weekdays, holidays, and all (total) days, respectively.

Models	MAPE_W	MAPE_H	MAPE_T	CVRMSE_W	CVRMSE_H	CVRMSE_T	NMAE_W	NMAE_H	NMAE_T
RF	8.71	10.64	9.32	12.03	15.42	12.91	8.84	11.04	9.41
GBM	8.68	10.79	9.35	11.97	15.45	12.88	8.78	11.09	9.38
XGBoost	8.85	10.68	9.44	12.12	15.31	12.96	8.95	11.07	9.50
LightGBM	8.62	10.51	9.23	11.79	15.06	12.65	8.71	10.81	9.26
CatBoost	8.64	10.75	9.31	11.94	15.63	12.90	8.75	11.10	9.36

Table 4
Performance comparison of tree-based ensemble models trained with internal factors for the educational building dataset. W, H, and T indicate weekdays, holidays, and all (total) days, respectively.

Models	MAPE_W	MAPE_H	MAPE_T	CVRMSE_W	CVRMSE_H	CVRMSE_T	NMAE_W	NMAE_H	NMAE_T
RF	5.03	7.40	5.79	7.40	11.58	8.52	5.07	7.57	5.72
GBM	4.92	7.13	5.63	7.16	11.31	8.27	4.94	7.30	5.56
XGBoost	4.91	7.04	5.59	7.16	11.16	8.23	4.93	7.20	5.53
LightGBM	4.88	6.92	5.53	7.22	11.12	8.26	4.92	7.09	5.49
CatBoost	4.91	7.00	5.58	7.26	11.26	8.33	4.94	7.18	5.53

Table 5
Performance comparison of tree-based ensemble models trained with external and internal factors for the educational building dataset. W, H, and T indicate weekdays, holidays, and all (total) days, respectively.

Models	MAPE_W	MAPE_H	MAPE_T	CVRMSE_W	CVRMSE_H	CVRMSE_T	NMAE_W	NMAE_H	NMAE_T
RF	4.48	6.50	5.13	6.56	10.69	7.68	4.63	6.75	5.19
GBM	4.44	6.43	5.08	6.49	10.59	7.60	4.56	6.66	5.11
XGBoost	4.41	6.39	5.04	6.41	10.61	7.55	4.52	6.64	5.07
LightGBM	4.44	6.33	5.04	6.41	10.29	7.45	4.54	6.55	5.07
CatBoost	4.41	6.54	5.09	6.43	10.70	7.59	4.54	6.78	5.12

Performance comparison

Tables 3 to 5 compare MAPE, CVRMSE, and NMAE for different input variable configurations by date type using the educational building dataset. When comparing MAPE, CVRMSE, and NMAE by different input variable configurations, the MAPE, CVRMSE, and NMAE values considering external and internal factors are the lowest. When comparing the performance of weekdays and holidays in terms of STL for educational buildings, weekdays performed better on average than holidays. Table 5 indicates that the LightGBM model considering external and internal factors derived the best prediction performance among most DT-based ensemble learning models; hence, it was chosen as the best STL model for the educational building dataset.

In addition, to analyze the influence of the size of the training set on forecasting accuracy, we set several training sets for the educational building dataset by considering the period by year, as follows:

- March 1, 2018, to February 28, 2019 (1 year),
- March 1, 2017, to February 28, 2019 (2 years),
- March 1, 2016, to February 28, 2019 (3 years).

Tables 6 to 8 present the MAPE, CVRMSE, and NMAE comparisons for different input variable configurations with different training set periods. We found that a longer training set period results in finer but better prediction results.

Model explanation

The explainability method has received considerable attention as a strategy to ensure model reliability. We employed the tree SHAP method to improve the explainability of the DT-based ensemble models by computing the contribution of each input variable to the prediction. We used a summary plot, dependent plot, heatmap plot, and other methods.

For the educational building dataset, we displayed how much each

Table 6

Performance comparison of tree-based ensemble models trained with one year of data for the educational building dataset. W, H, and T indicate weekdays, holidays, and all (total) days, respectively.

Models	MAPE_W	MAPE_H	MAPE_T	CVRMSE_W	CVRMSE_H	CVRMSE_T	NMAE_W	NMAE_H	NMAE_T
RF	4.68	6.98	5.41	6.92	11.39	8.13	4.76	7.10	5.37
GBM	4.69	6.82	5.37	6.90	11.17	8.05	4.79	6.96	5.36
XGBoost	4.71	6.78	5.37	6.93	11.07	8.04	4.84	6.99	5.40
LightGBM	4.66	6.89	5.37	6.83	11.16	8.00	4.76	7.04	5.36
CatBoost	4.61	6.97	5.36	6.78	11.47	8.06	4.73	7.12	5.35

Table 7

Performance comparison of tree-based ensemble models trained with two years of data for the educational building dataset. W, H, and T indicate weekdays, holidays, and all (total) days, respectively.

Models	MAPE_W	MAPE_H	MAPE_T	CVRMSE_W	CVRMSE_H	CVRMSE_T	NMAE_W	NMAE_H	NMAE_T
RF	4.53	6.57	5.18	6.69	10.96	7.85	4.67	6.75	5.21
GBM	4.48	6.50	5.12	6.55	10.79	7.70	4.56	6.66	5.11
XGBoost	4.42	6.41	5.06	6.47	10.69	7.62	4.51	6.60	5.06
LightGBM	4.49	6.48	5.13	6.57	10.53	7.64	4.59	6.62	5.12
CatBoost	4.41	6.54	5.09	6.47	10.87	7.67	4.52	6.71	5.09

Table 8

Performance comparison of tree-based ensemble models trained with three years of data for the educational building dataset. W, H, and T indicate weekdays, holidays, and all (total) days, respectively.

Models	MAPE_W	MAPE_H	MAPE_T	CVRMSE_W	CVRMSE_H	CVRMSE_T	NMAE_W	NMAE_H	NMAE_T
RF	4.49	6.41	5.10	6.60	10.53	7.65	4.64	6.62	5.16
GBM	4.47	6.46	5.11	6.54	10.58	7.63	4.59	6.67	5.13
XGBoost	4.40	6.35	5.03	6.43	10.51	7.53	4.52	6.56	5.06
LightGBM	4.48	6.41	5.10	6.55	10.36	7.57	4.61	6.61	5.13
CatBoost	4.46	6.48	5.11	6.56	10.62	7.66	4.60	6.67	5.14

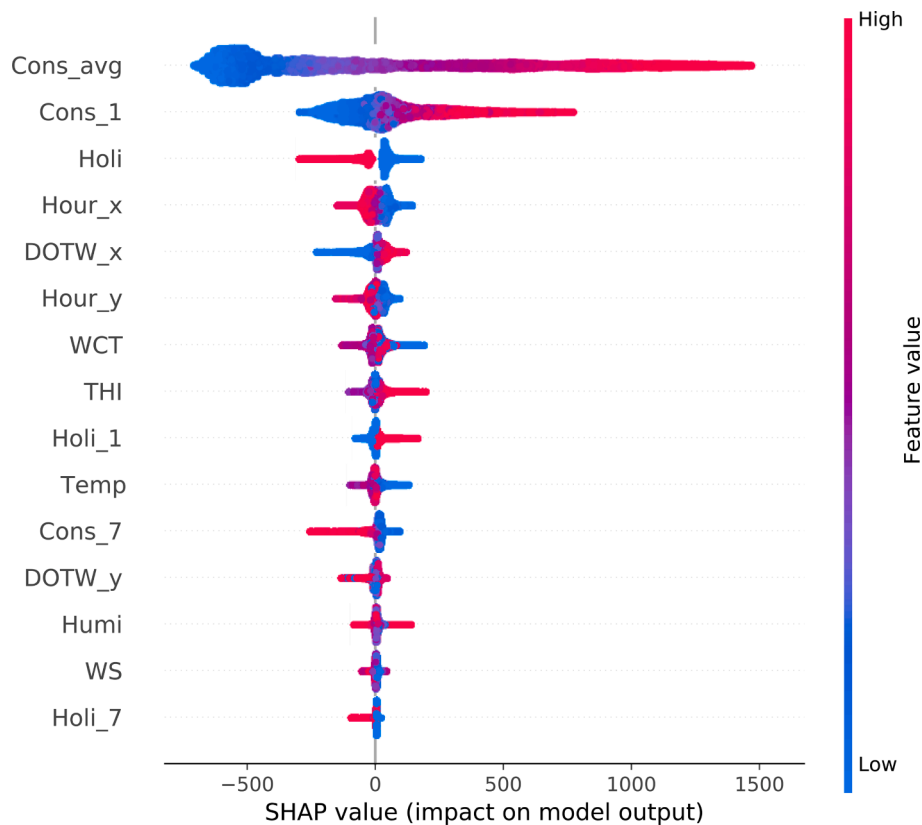


Fig. 2. Summary plot of the LightGBM model for the educational building dataset (see Table 1 for the definition of abbreviations).

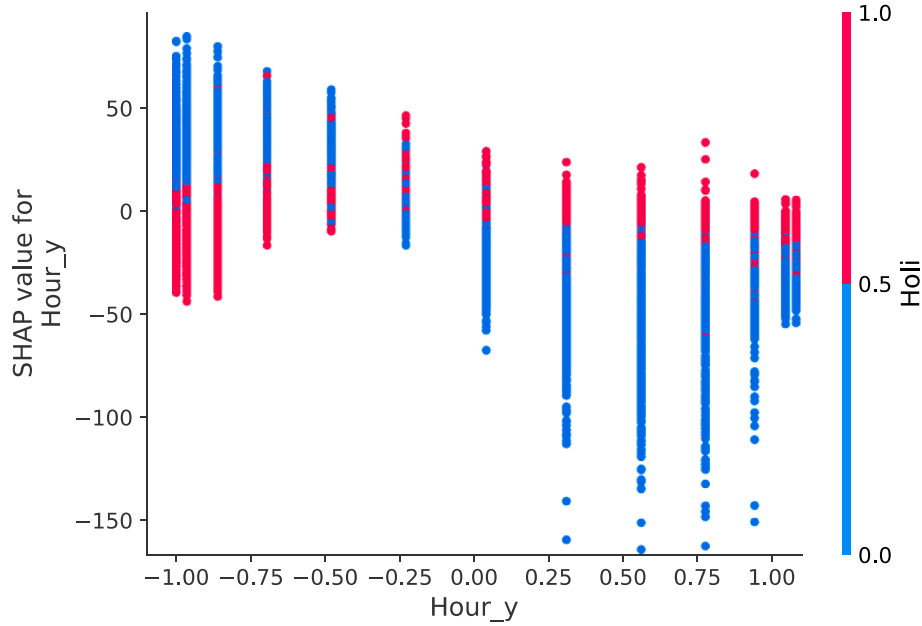


Fig. 3. PDP for the SHAP value distribution and interaction between *Hour_y* and the holiday indicator (*Holi*).

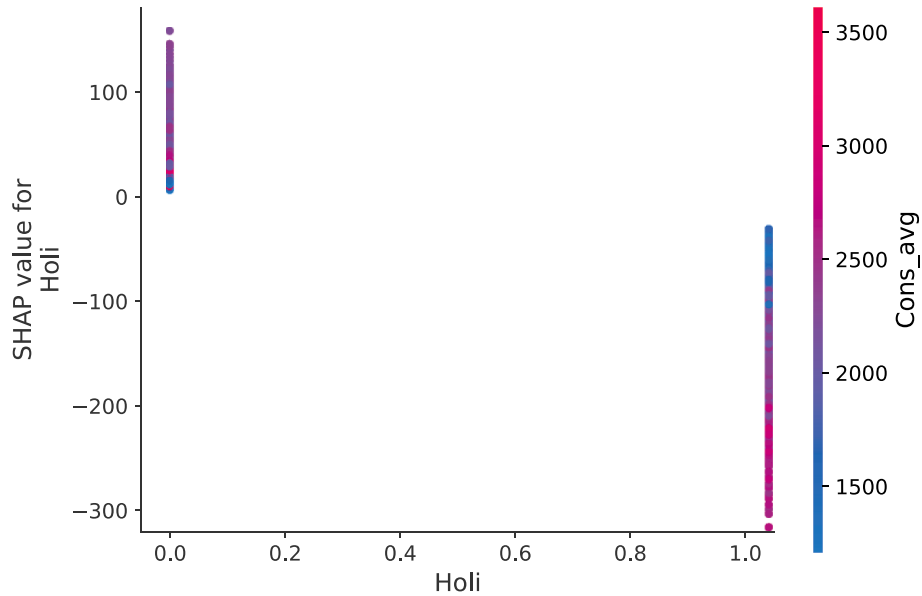


Fig. 4. PDP for the SHAP value distribution and interaction of holiday indicator (*Holi*) corresponding to weekly average electricity consumption (*Cons_avg*).

attribute (a.k.a. input variable) features in the LightGBM model in Figs. 2 to 7. A summary plot exhibits the input variables in order of greatest influence, and a larger feature value has a greater influence on the forecasting model. Fig. 2 is a summary plot of the LightGBM trained for external and internal factors. We confirmed that the weekly average electricity consumption (*Cons_avg*) and electricity consumption one day prior (*Cons₁*) substantially influence the LightGBM model construction. In weather information, the THI and WCT have more influence on the LightGBM model construction than the temperature, humidity, and wind speed.

A PDP is a scatter plot that represents the relationship between one or two features and samples. Each point represents a sample, and the corresponding SHAP values represent the contribution to the sample. A PDP plots the feature value on the x-axis and the SHAP value of the same feature on the y-axis. The vertical dispersion of the data points represents the influences of the interaction. Fig. 3 is a PDP for the SHAP value

distribution and interaction between *Hour_y* and the holiday indicator (*Holi*). Additionally, -1 and 1 indicate noon and midnight, respectively, in the value of *Hour_y*. We confirmed that the influence of the SHAP value increased positively around daytime and negatively around the night on weekdays. In contrast, the influence of the SHAP value increases negatively around daytime and positively around the night on holidays.

Fig. 4 is a PDP for the SHAP value distribution and interaction of the holiday indicator (*Holi*), corresponding to the weekly average electricity consumption (*Cons_avg*). We confirmed that the influence of the holiday indicator on the LightGBM model increases positively when the dates in the sample are weekdays. In contrast, it increases negatively when the dates in the sample are holidays, and the absolute SHAP values are larger when it was a holiday than when it was a weekday.

Figs. 5 and 6 are PDPs for the SHAP value distribution and interaction of the THI and WCT corresponding to the weekly average electricity consumption, respectively. In these figures, we confirmed that the SHAP

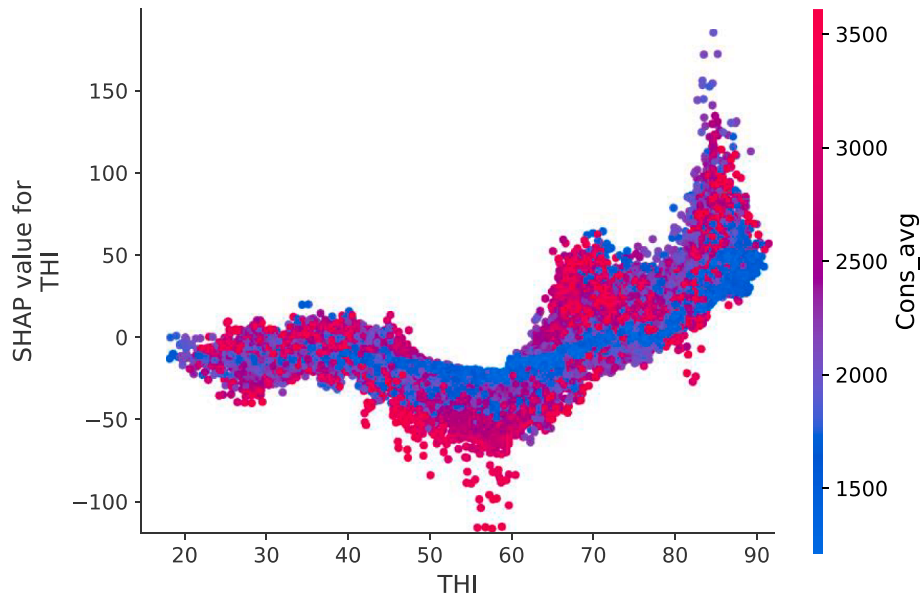


Fig. 5. PDP for the SHAP value distribution and interaction of temperature-humidity index (THI) corresponding to the weekly average electricity consumption.

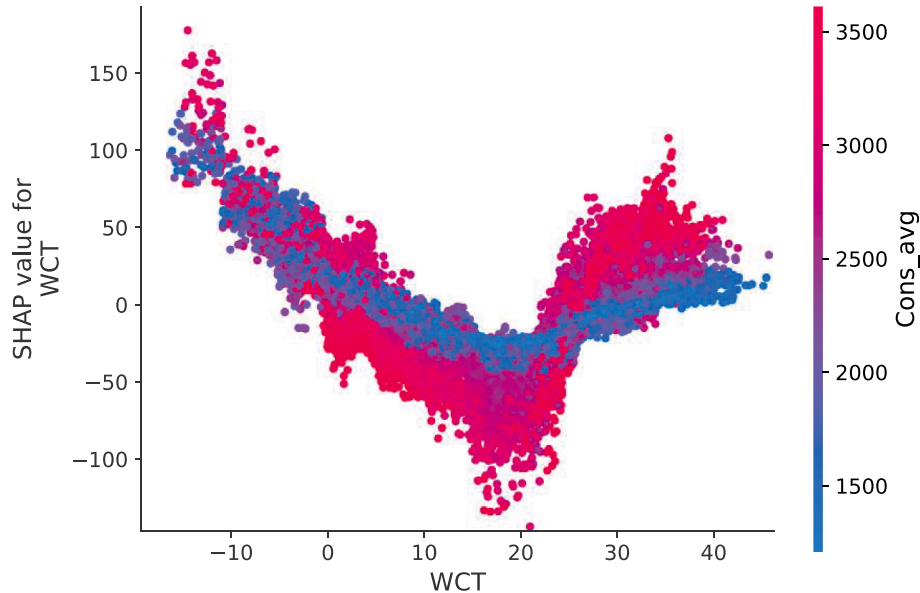


Fig. 6. PDP for the SHAP value distribution and interaction of wind chill temperature (WCT) corresponding to the weekly average electricity consumption.

values of THI and WCT increase when they are high and low, respectively. We can estimate that electricity consumption increases when the THI falls below a certain value (i.e., in winter) in Fig. 5. We also can estimate that the electricity consumption increases when the WCT rises above a certain value (i.e., in summer) in Fig. 6.

Fig. 7 is a heatmap plot for passing a matrix of SHAP values. The instances are plotted on the x-axis, and the input variables are plotted on the y-axis. The SHAP values are encoded on a color scale. As a result, samples with the same model output for the same reason are clustered together by ordering the samples based on their explanation similarity in hierarchical clustering. The LightGBM model output is presented above the heatmap matrix, with the global importance of each input variable displayed as a bar plot on the right side of this figure. We confirmed that the values of $Cons_{avg}$ and $Cons_1$ are high when the value of $f(x)$ is high; hence, historical electricity consumption variables have the most significant influence on the STLTF in an educational building.

Discussion

Although the proposed method could perform accurate and explainable hourly electricity consumption forecasting, it is challenging to adopt this method in the energy sector if it does not yield a better prediction performance than the existing popular STLTF models. To verify the superiority of the proposed method, we implemented diverse existing STLTF models on the educational building dataset because this dataset contains a sufficient data collection period of five years. The existing STLTF models include Tao's vanilla benchmark model, Huo's SVR and RF models, Nemeth's XGBoost and LightGBM models, and Park's DNN model [66–69]. Tao's vanilla benchmark model, which is based on MLR, is a commonly used regression model for STLTF. It was used as the benchmark model in the Global Energy Forecasting Competition 2012 (GECCom2012) [66,70].

Huo's SVR and RF models demonstrated satisfactory performance in publicly available datasets, such as the European Network on Intelligent

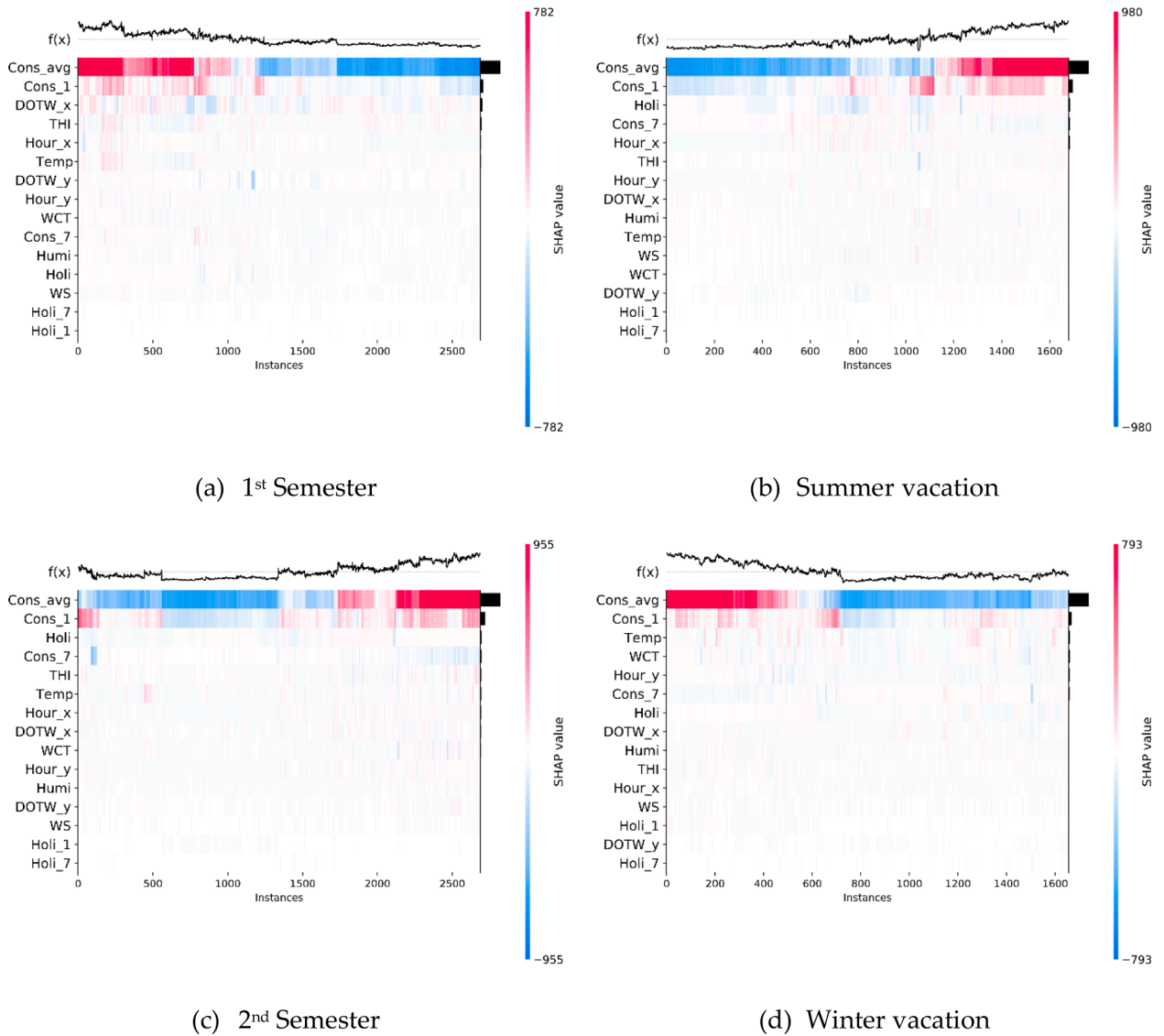


Fig. 7. Heatmap plot of the LightGBM model for the educational building dataset (see Table 1 for the definition of abbreviations).

Technologies (EUNITE) and the New York Independent System Operator (NYISO) datasets [67]. Because the implementation environment they considered (i.e., input variable, model construction, etc.) could be experimented on in the same way in this study, we determined these models to be baseline models. Nemeth's XGBoost and LightGBM models performed satisfactorily for thermal power output [68]. For model training, they considered only timestamp information, such as the year,

hour, day of the week, and holiday indicator. We determined these models to be baseline models to compare the performance when only timestamp information was used. We implemented the input variables of Huo's SVR and RF and Memeth's XGBoost and LightGBM in the same way as in [67,68] and constructed these models by determining the optimal hyperparameter values via a grid search.

Park's DNN model, our prior study, was developed to predict hourly

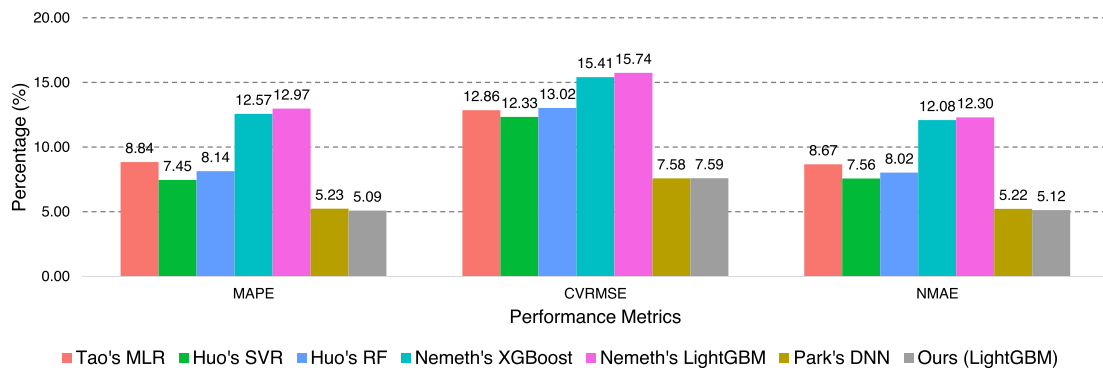


Fig. 8. Comparison of prediction performance through MAPE, CVRMSE, and NMAE.

electricity consumption for an industrial building [69]. We partitioned the dataset into training, validation, and testing sets and set up input variables for model training. We built RF and XGBoost models on the training set and forecasted electricity consumption on the validation set in the first stage. In the second stage, we built a DNN model using existing input variables and prediction values from the RF and XGBoost models from the validation set and forecasted electric load on the testing set. Park's DNN model outperformed single machine learning and existing STLF models [32,45]. We implemented the input variables in the same way as this study and divided the dataset into training, validation, and testing sets from March 8, 2015, to February 28, 2018; March 1, 2018, to February 28, 2019; and March 1, 2019, to February 29, 2020, respectively. We constructed the RF, XGBoost, and DNN models by determining the optimal hyperparameter values via a grid search for Park's DNN model construction.

Fig. 8 presents diverse STLF model prediction performances over the entire period of the testing set. In this figure, the LightGBM model yielded the best prediction performance for the MAPE and NMAE in the educational building dataset. Park's DNN model yielded a similar prediction performance as the LightGBM model and yielded the best prediction performance for CVRMSE; however, it was time-consuming despite using a GPU. In this regard, the LightGBM model is more competitive than Park's DNN model. Nemeth's XGBoost and LightGBM models, which were trained with only timestamp information, exhibited unsatisfactory performance because it was difficult to sufficiently train for complex building electricity consumption patterns. Hence, we concluded that the LightGBM model could sufficiently be applicable in EMSs because it provides explainable electricity consumption forecasting with a more stable prediction performance.

Conclusions

In this paper, we proposed the XELF methodology for educational buildings. We first collected and preprocessed data for the input variable configuration, considering external and internal factors. Then, we divided the datasets into training and testing sets and constructed five DT-based ensemble learning models (RF, GBM, XGBoost, LightGBM, and CatBoost) using the training set. Using the testing set, we compared their prediction performance for the MAPE, CVRMSE, and NMAE. The LightGBM trained with external and internal factors achieved better prediction performance than the other forecasting models in the educational building dataset. We applied the SHAP to explain the LightGBM model in the educational building dataset.

We confirmed that THI and WCT exhibited more influence on forecasting model construction than temperature, humidity, and wind speed in weather information. Because the SHAP value was large for THI in summer and WCT in winter, we estimated that they contribute to more accurate day-ahead hourly electricity consumption forecasting for summer and winter, respectively. In the educational building dataset, the weekly average electricity consumption and electricity consumption one day prior exhibited more influence on the LightGBM model construction than the other input variables. A larger training dataset and longer training period for modeling lead to better accuracy.

One of the main limitations of this study is that the analysis was performed for only hourly electricity consumption forecasting from one dataset. Important information could be discovered when analyzing other building types, such as commercial (office) and industrial, and other time intervals, such as 15- and 10-minute intervals, for very STLF. In addition, explainable hybrid forecasting models based on two or more single AI techniques or explainable online learning models that can learn to reflect recent electricity consumption trends and patterns should be developed and described for more accurate energy forecasting. The XELF model did not perform multistep-ahead or probabilistic forecasting in this study. The XELF model must be improved to perform multistep-ahead and probabilistic forecasting for better energy services.

We plan to collect more energy-related data during the coronavirus

disease 2019 (COVID-19) period to analyze the correlation between COVID-19 and electricity consumption in-depth using XELF-based behavioral analysis, which could be important in future pandemics. We will apply the XELF methodology to solar irradiance and photovoltaic prediction to contribute to sustainable development. We also plan to develop an advanced STLF model, such as a multistep-ahead forecasting model or a probabilistic forecasting model, and present an architecture that can explain the advanced STLF model. In addition, we plan to study how to apply an STLF model that considers the common time-series correlation characteristics of energy demand and generation. We plan to develop an electricity energy management and trading system based on demand and supply forecasting that considers information on external factors, such as user consumption behavioral patterns. Finally, selecting features without using all input variables is important for quick model training depending on the computing environment. We will develop an XAI methodology to remove unnecessary variables by analyzing the STLF model and training with only meaningful variables for high prediction performance with a low computing cost.

CRedit authorship contribution statement

Jihoon Moon: Conceptualization, Formal analysis, Methodology, Software, Visualization, Writing – original draft. **Seungmin Rho:** Investigation, Supervision, Validation, Writing – review & editing. **Sung Wook Baik:** Data curation, Funding acquisition, Project administration, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The dataset used to support the findings of this study are included within the [Supplementary Material \(https://doi.org/10.1016/j.seta.2022.102888\)](https://doi.org/10.1016/j.seta.2022.102888). Due to university privacy, we performed min-max normalization for electricity consumption data, namely Cons_7, Cons_1, Cons_avg, and Consumption in the dataset.

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